

Worksheet — Buoyancy Simulation

Use with *Simulation - Buoyancy.html* • goes with Station 4 (Archimedes' principle)

Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open biba0112.github.io/physics-sims/08-buoyancy in a browser (offline: *Simulation - Buoyancy.html*). Choose the object, its volume and the liquid, then lower it with the **d** slider. The overflow runs into the beaker on the balance. 🖱️ **Let go** releases the object.

Part 1 before the lab (≈ 15 min) • **Part 2** during the lab, Station 4 (≈ 10 min) • **Part 3** homework (≈ 30 min).

Use $g = 9.81 \text{ N kg}^{-1}$, $\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$. $1 \text{ cm}^3 = 1 \times 10^{-6} \text{ m}^3$.



Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. As the cylinder goes into the water, what happens to the newton meter reading?			
2. Fully under water, you lower it further (not touching the bottom). Reading?			
3. Compare the buoyancy force with the weight of the water collected.			
4. You lower a pine block ($\rho = 500 \text{ kg m}^{-3}$). What happens?			

5. Draw the free-body diagram of the cylinder hanging fully under water, then calculate the buoyancy force on the 63 cm^3 cylinder.

Free-body diagram and working

Part 2 — During the lab: Station 4

	Weight in air / N	Apparent weight in water / N	F_b / N	Displaced water / kg	its weight / N
Simulation					
Our measurement					

6. Does your data support Archimedes' principle? Give one reason why the two values in your own row are not exactly equal.

7. Why does a steel ship float while a steel nail sinks? Use your plasticine boat to explain.

Part 3 — Homework

8. Record at least 5 fully-submerged combinations (objects, volumes, liquids). Plot F_b against the weight of liquid displaced. Gradient = _____

Table of results

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9. Mystery stone: weight in air = _____ N, in water = _____ N. Calculate its density.

Working

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10. What fraction of a floating ice cube ($\rho = 920 \text{ kg m}^{-3}$) is under water? What fraction of a cork ($\rho = 240 \text{ kg m}^{-3}$) in saturated salt water ($\rho = 1200 \text{ kg m}^{-3}$)?

Working

11. A steel block ($\rho = 7850 \text{ kg m}^{-3}$) is put in mercury ($\rho = 13\,600 \text{ kg m}^{-3}$). Does it float? Draw its free-body diagram and find the fraction submerged.

Working